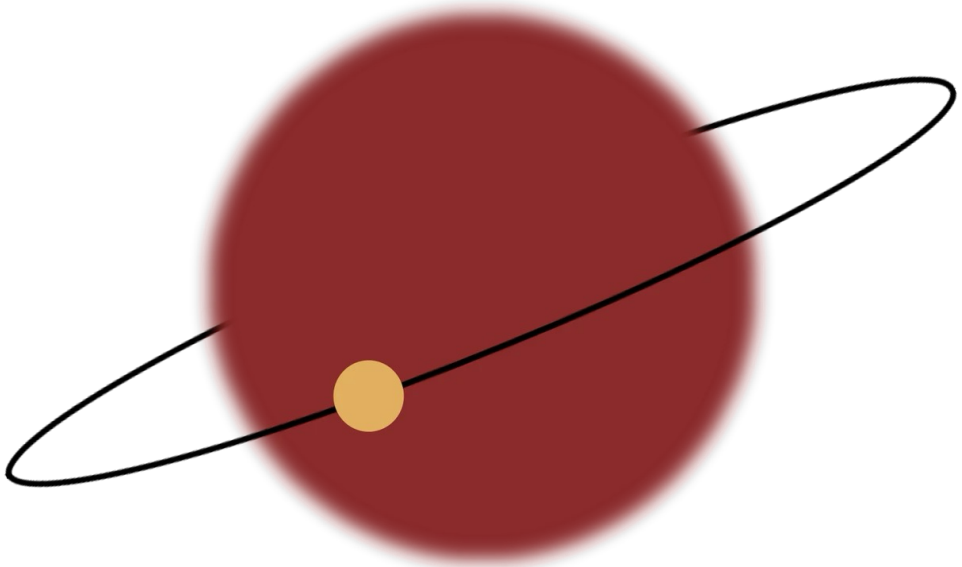
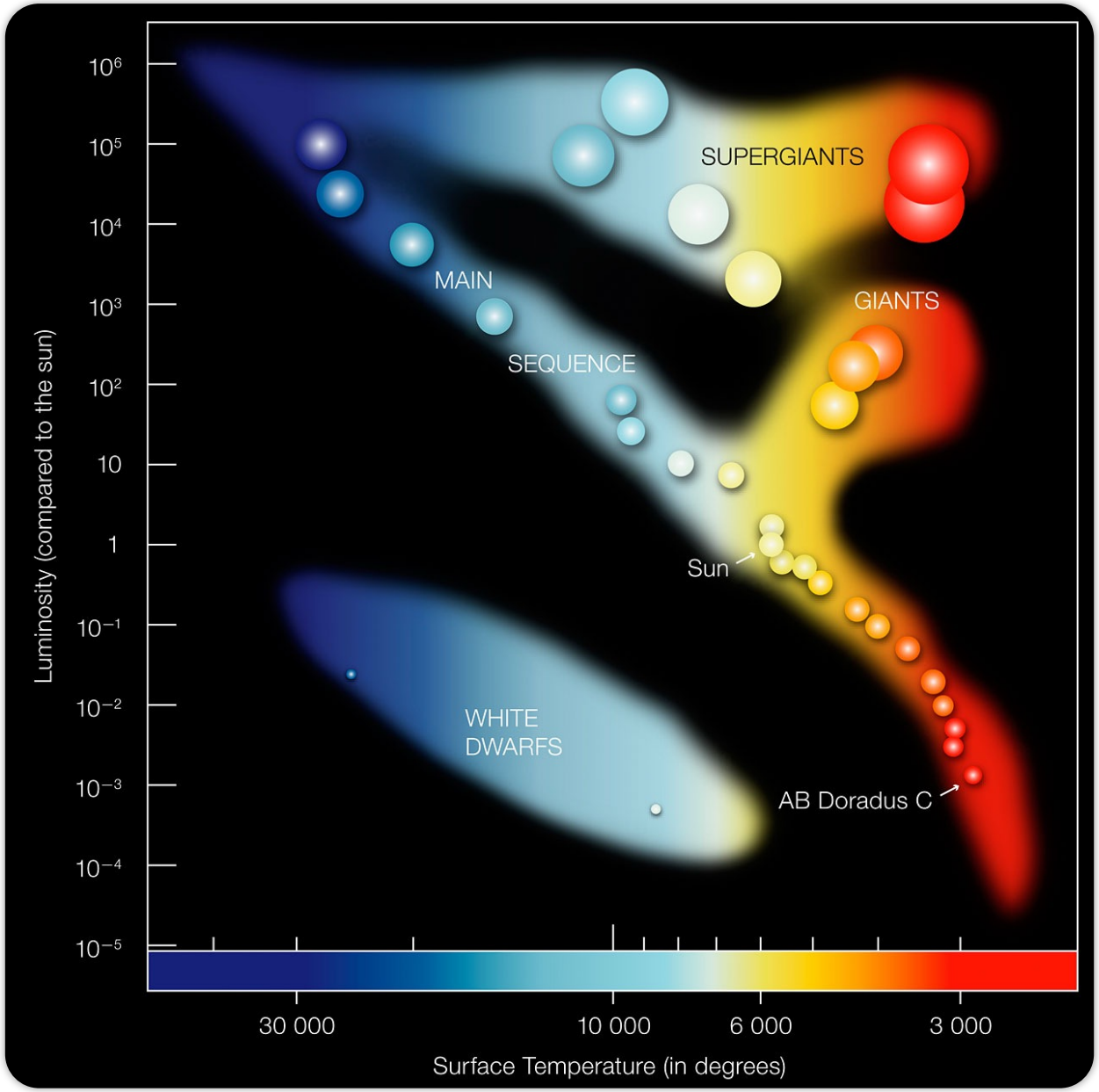


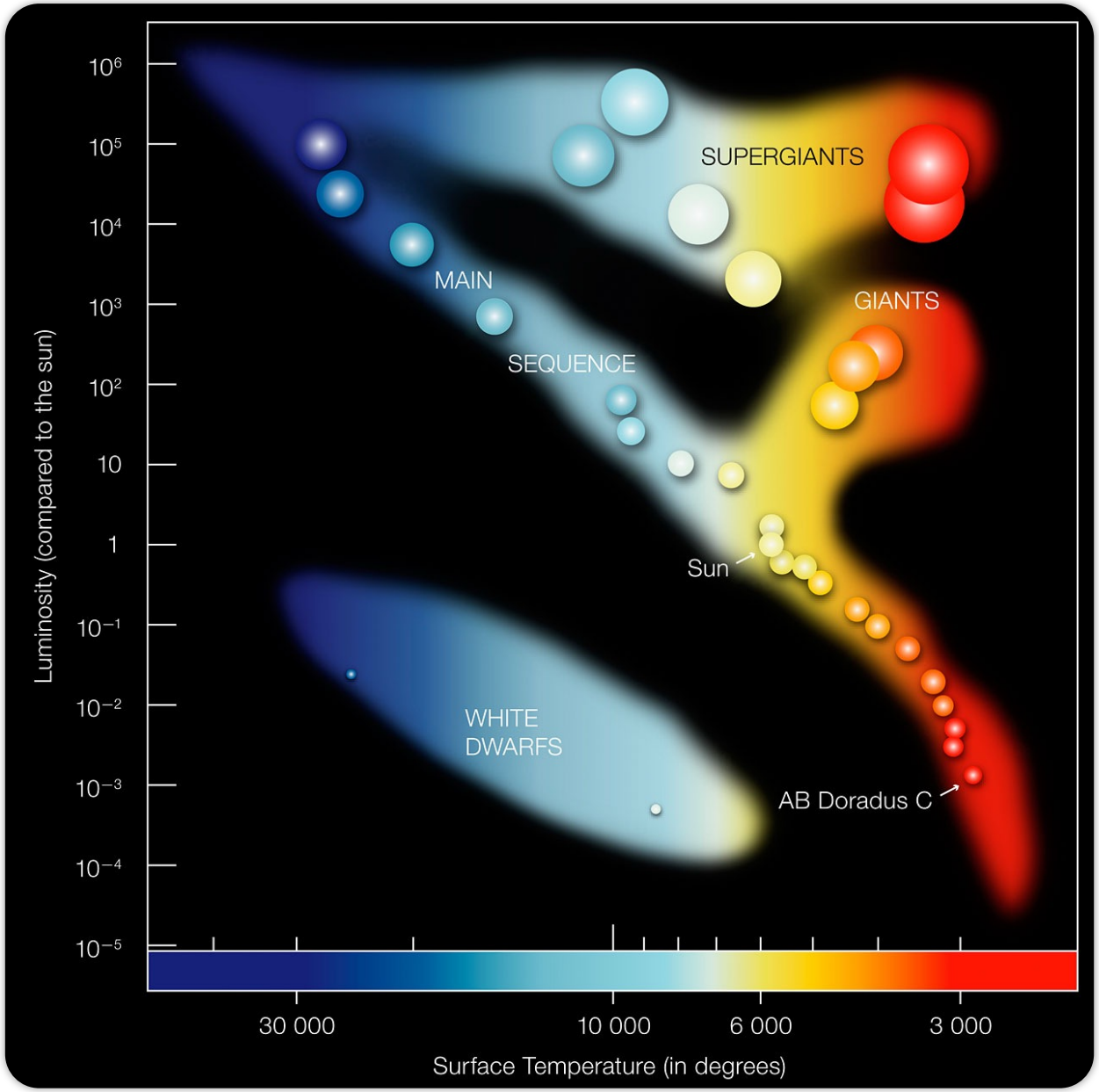
How to Find a Planet

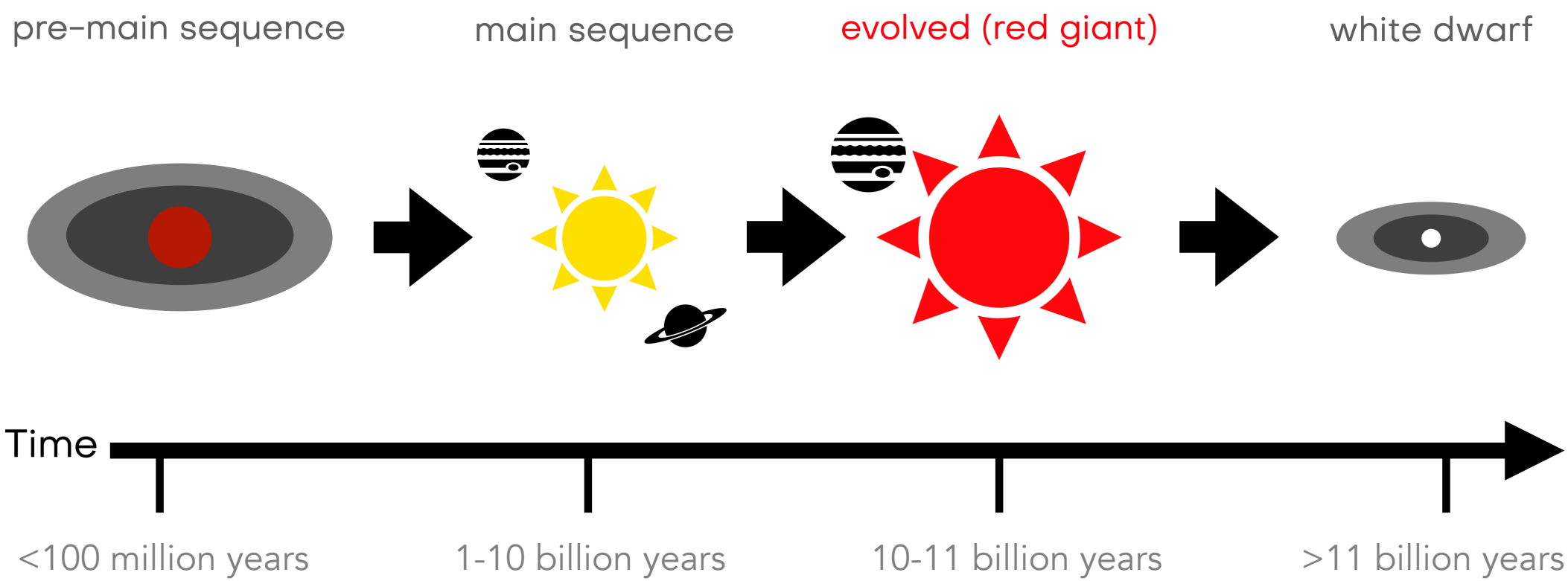
Dr. Nicholas Saunders
ASTR 2550, Yale University
16 October 2026

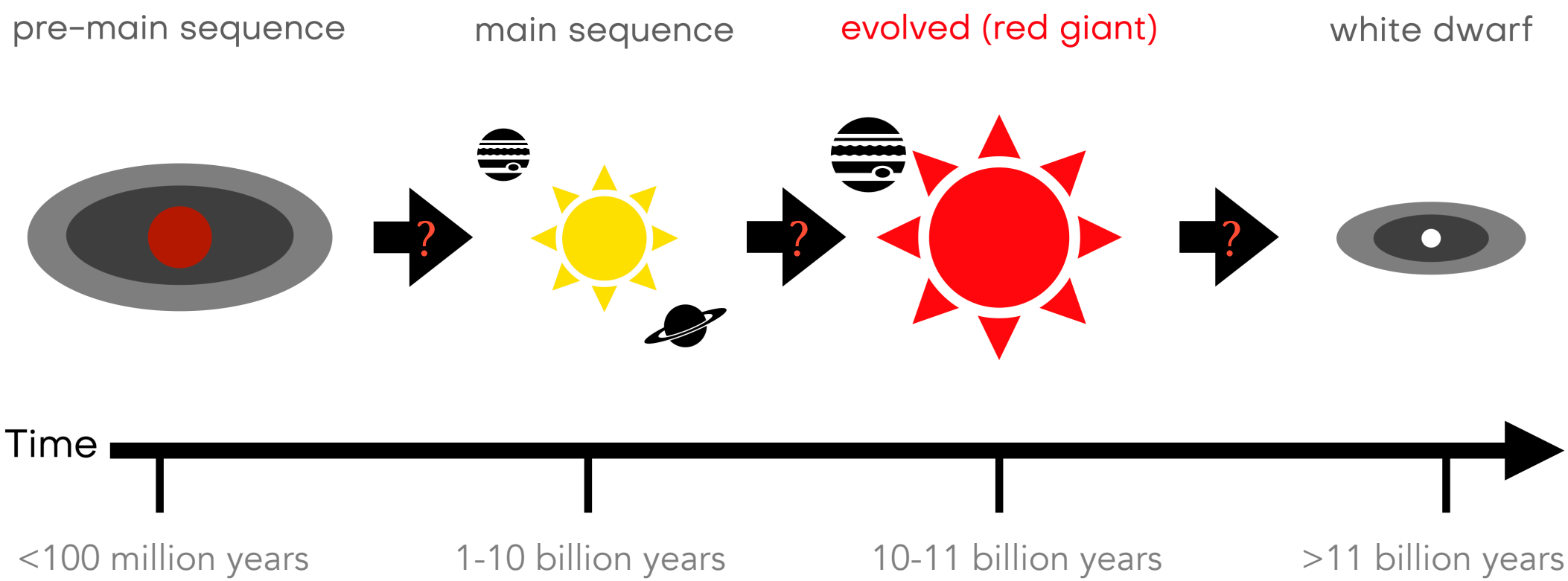
Planets have been discovered orbiting stars with a diverse range of masses, temperatures, and ages.



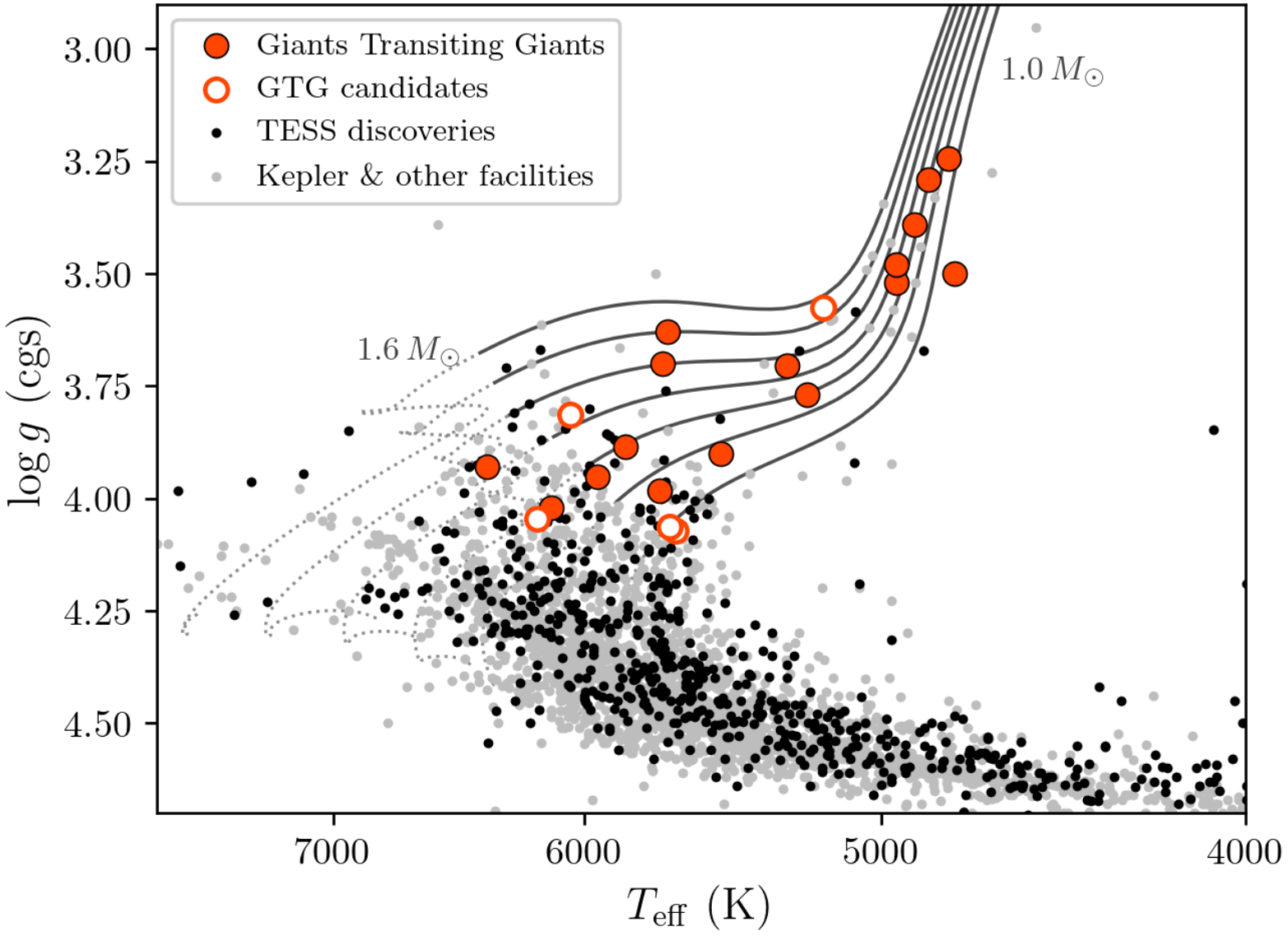
As stars like the sun near the ends of their lives, they grow larger, cooler, and more luminous.




pre-main sequence
main sequence
evolved (red giant)
white dwarf
Time
<100 million years
1-10 billion years
10-11 billion years
>11 billion years

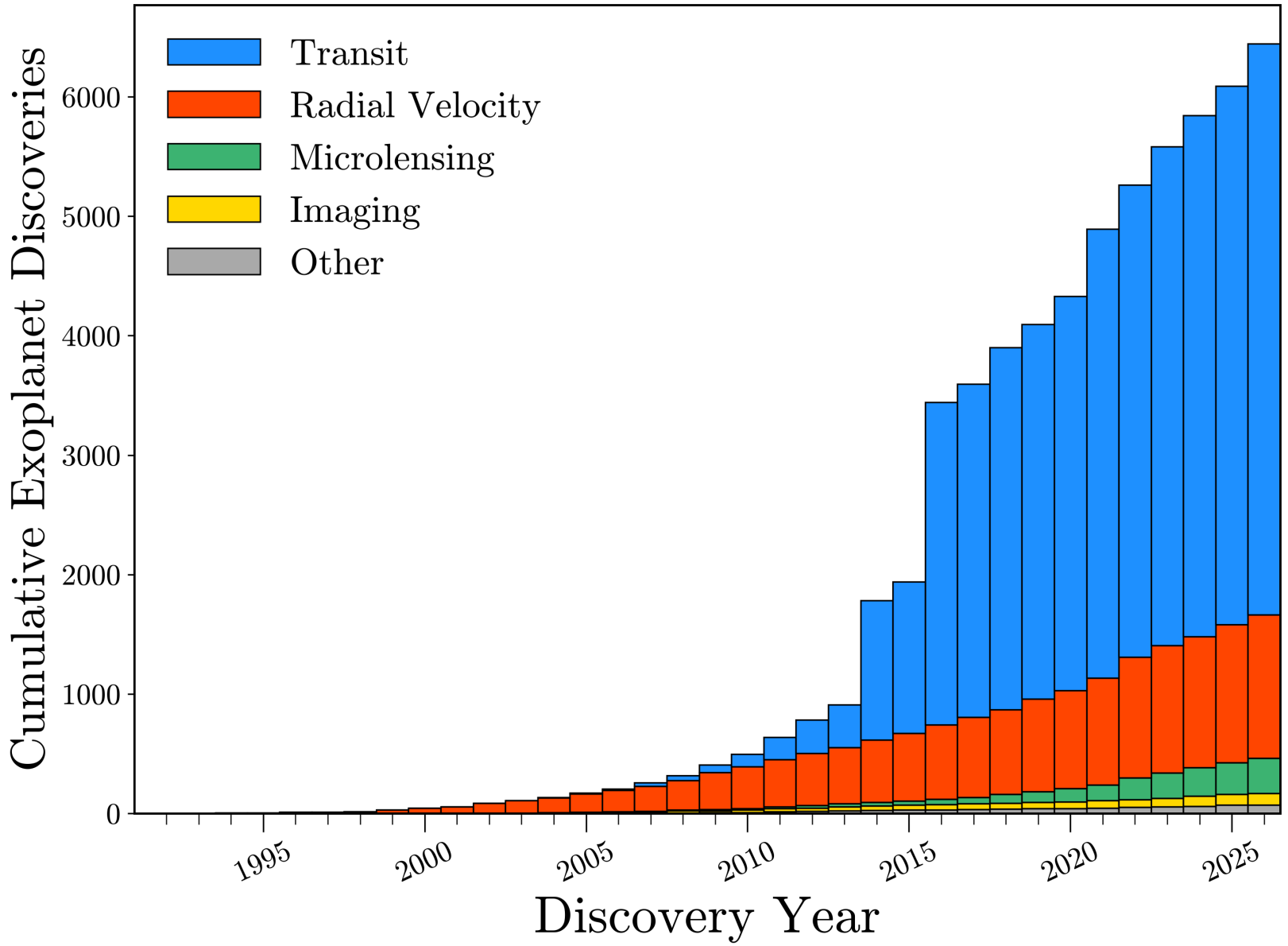

pre-main sequence
main sequence
evolved (red giant)
white dwarf
?
?
?
Time
<100 million years
1-10 billion years
10-11 billion years
>11 billion years

Our search is finding planets around the most evolved stars



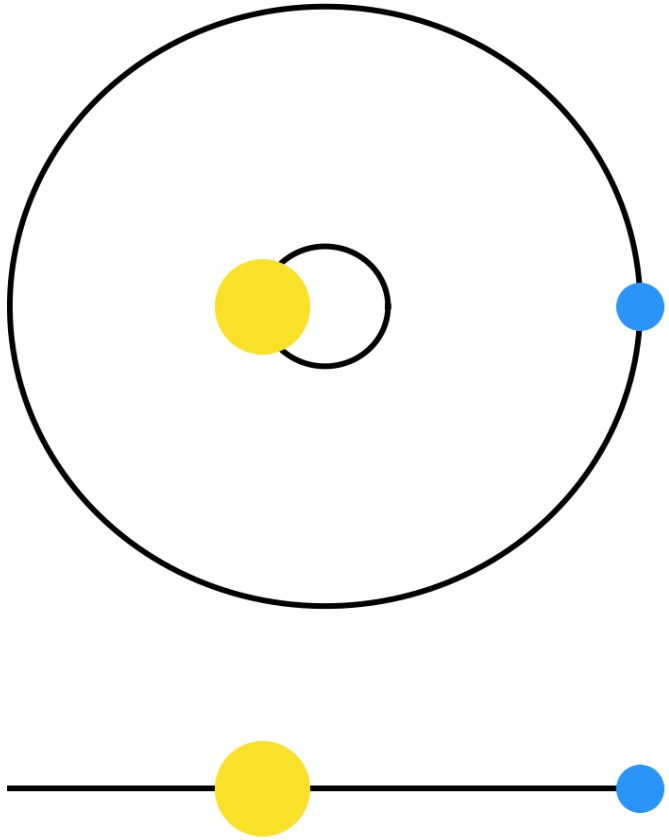
Giants Transiting Giants Papers:

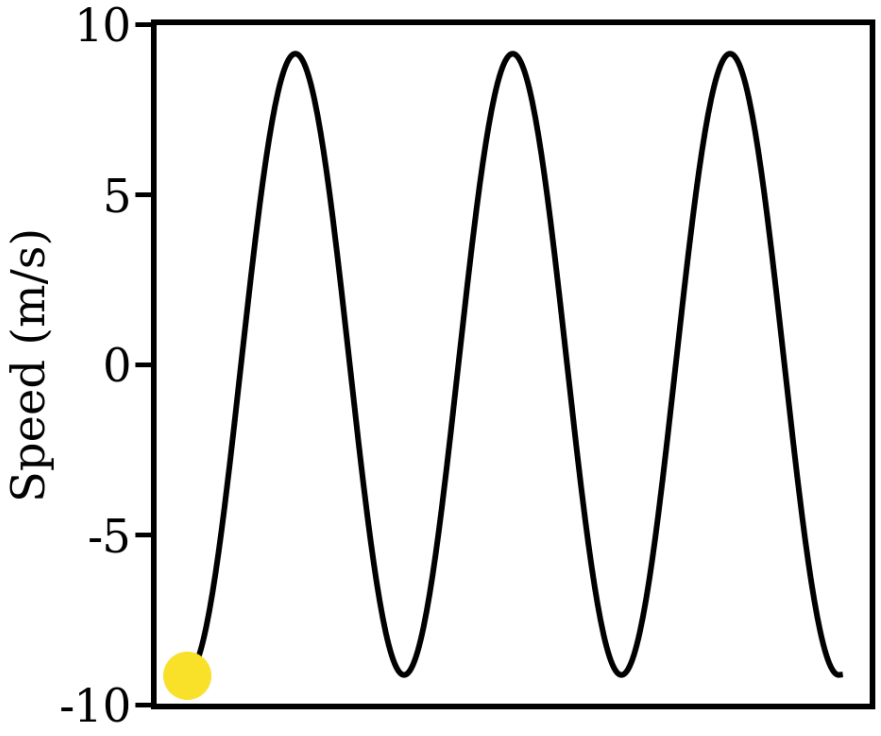
Saunders et al. (2022)
Grunblatt, ***Saunders*** et al. (2022)
Grunblatt, ***Saunders*** et al. (2023)
Pereira et al. incl. ***Saunders*** (2024)
Grunblatt, ***Saunders*** et al. (2024)
Saunders et al. (2024)
Saunders et al. (2025)
Page, ***Saunders*** et al. (in review)
Saunders et al. (in review)

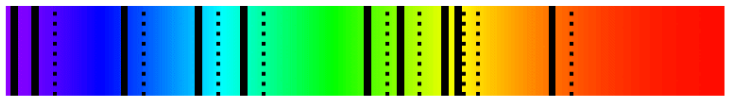

Transit
Radial Velocity
Microlensing
Imaging
Other
Cumulative Exoplanet Discoveries
0
1000
2000
3000
4000
5000
6000
1995
2000
2005
2010
2015
2020
2025
Discovery Year

The Radial Velocity Method

Alysa Obertas (@AstroAlysa)







The Radial Velocity Method

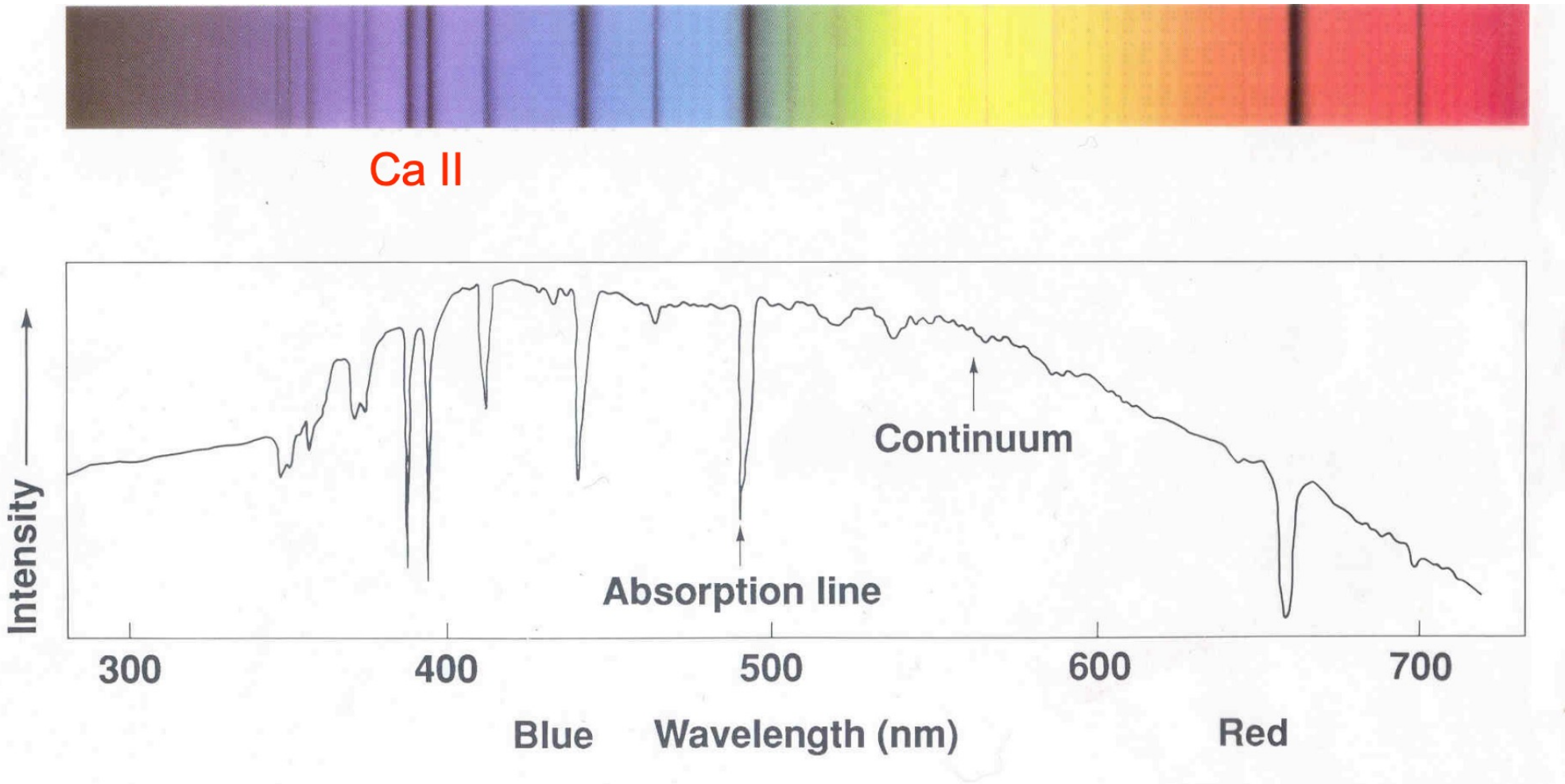
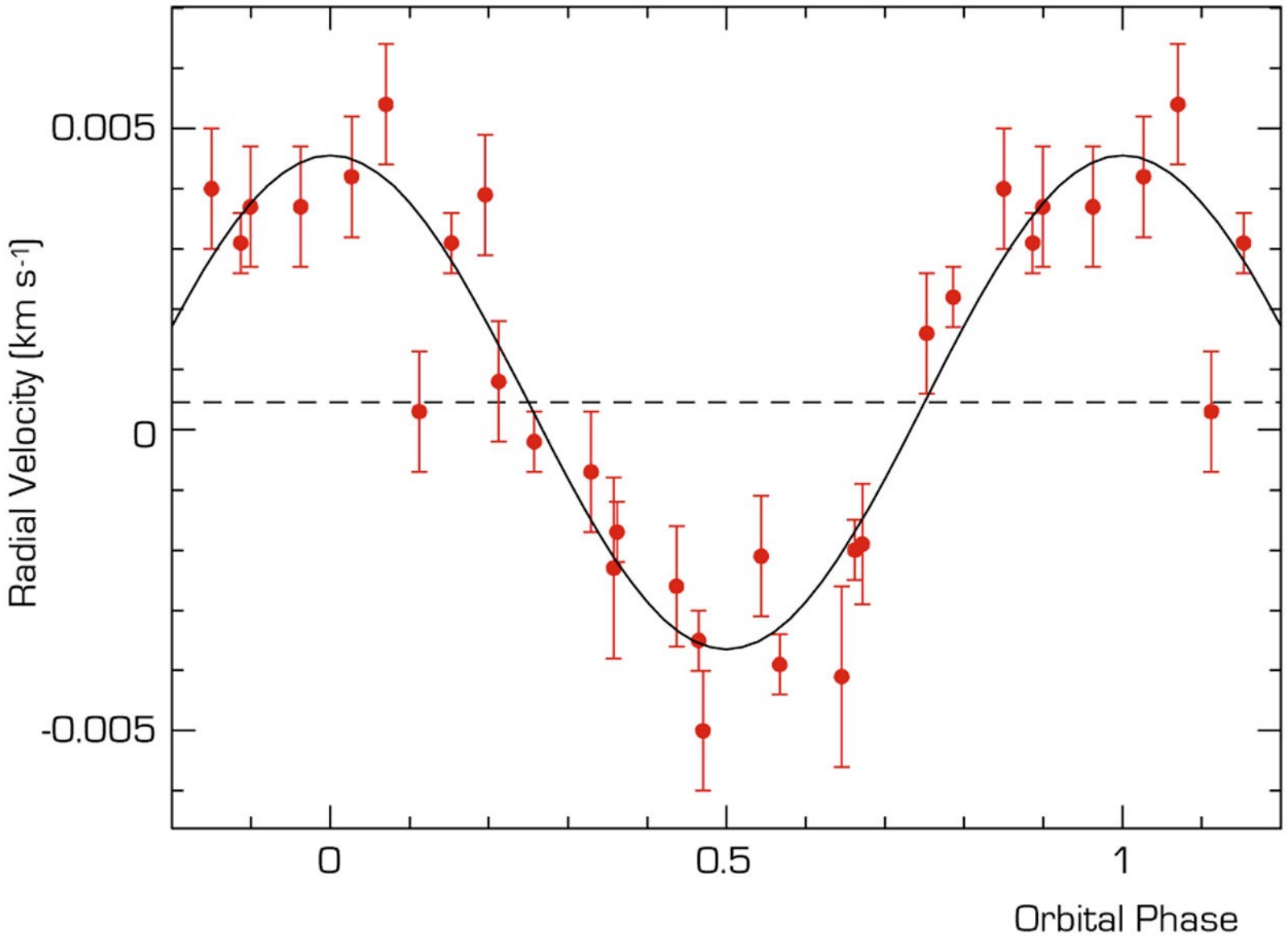


Image: Princeton

The Radial Velocity Method



$$K = \left(\frac{2\pi G}{M_\star^2 P}\right)^{1/3} \frac{1}{\sqrt{1-e^2}} M_p \sin i$$

- K — RV semi-amplitude
- G — gravitational constant
- $M_\star$ — stellar mass
- P — orbital period
- e — eccentricity
- M_p — planet mass
- i — orbital inclination

Image: ESO

The Radial Velocity Method

measuring shifts of spectral lines due to the star wobbling as the planet orbits

$$K = \left(\frac{2\pi G}{M_\star^2 P}\right)^{1/3} \frac{1}{\sqrt{1-e^2}} M_p \sin i$$

best at finding

small (terrestrial) **or** **large** (gas giant)

planets on

short (days) **or** **long** (years)

orbits.

The Radial Velocity Method

measuring shifts of spectral lines due to the star wobbling as the planet orbits

$$K = \left(\frac{2\pi G}{M_\star^2 P}\right)^{1/3} \frac{1}{\sqrt{1-e^2}} M_p \sin i$$

best at finding

small (terrestrial) **or** **large** (gas giant)

planets on

short (days) **or** **long** (years)

orbits.

The Radial Velocity Method

measuring shifts of spectral lines due to the star wobbling as the planet orbits

$$K = \left(\frac{2\pi G}{M_\star^2 P}\right)^{1/3} \frac{1}{\sqrt{1-e^2}} M_p \sin i$$

best at finding

small (terrestrial) **or** **large** (gas giant)

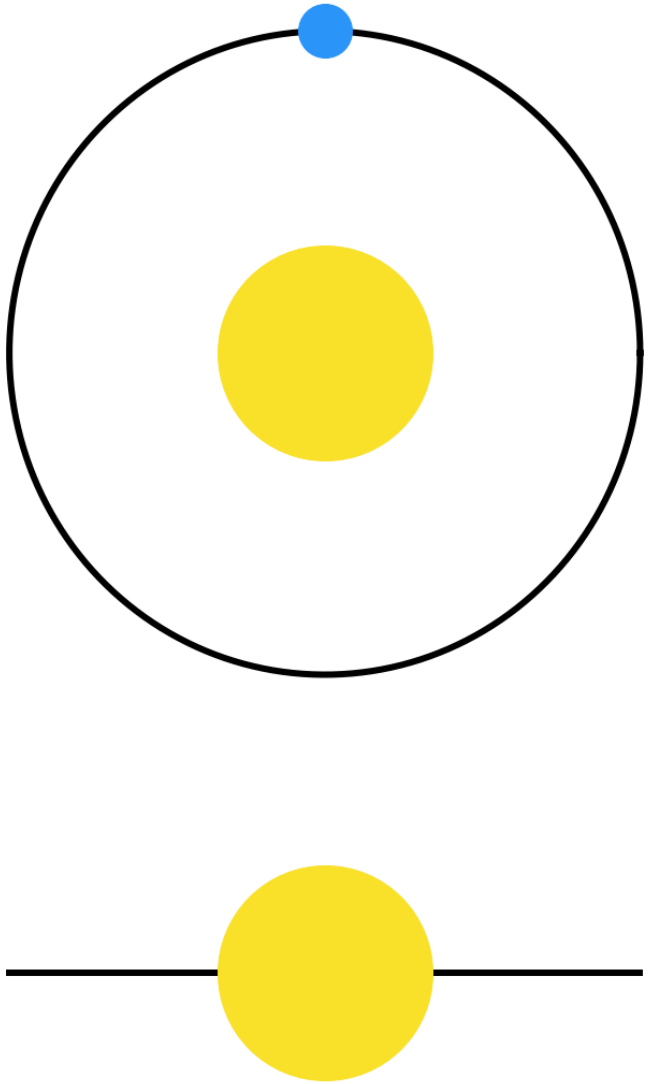
planets on

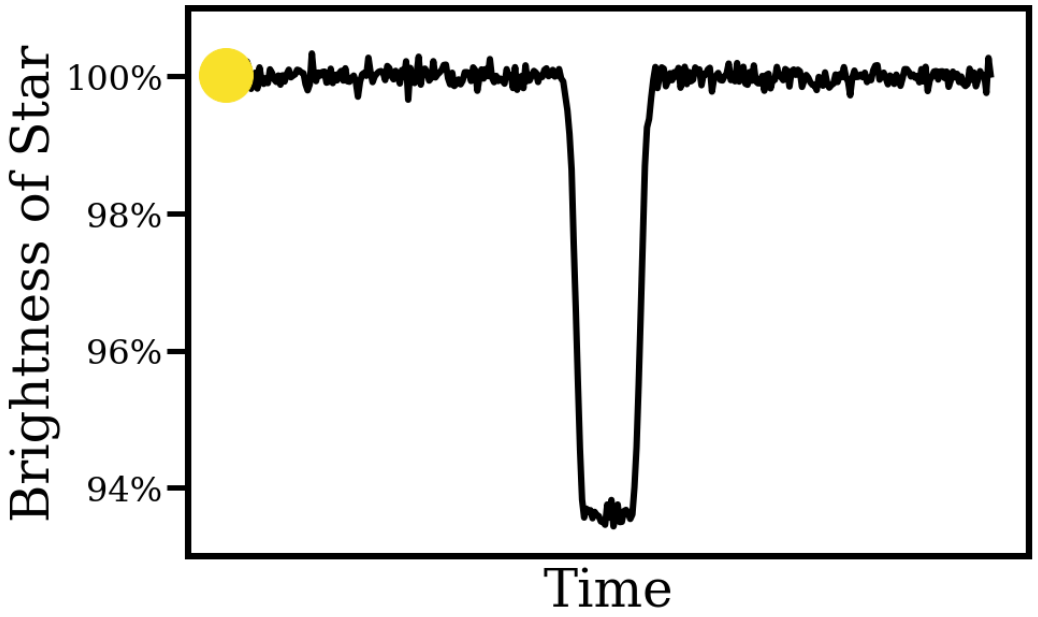
short (days) **or** **long** (years)

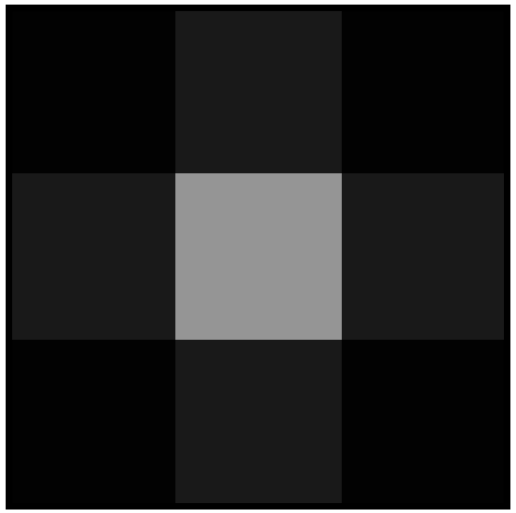
orbits.

The Transit Method

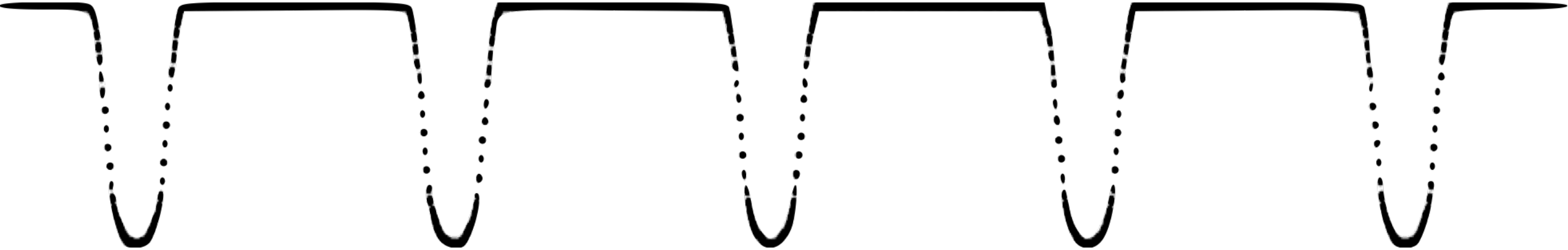
Alysa Obertas (@AstroAlysa)



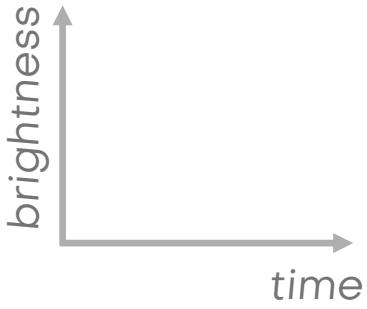




The Transit Method

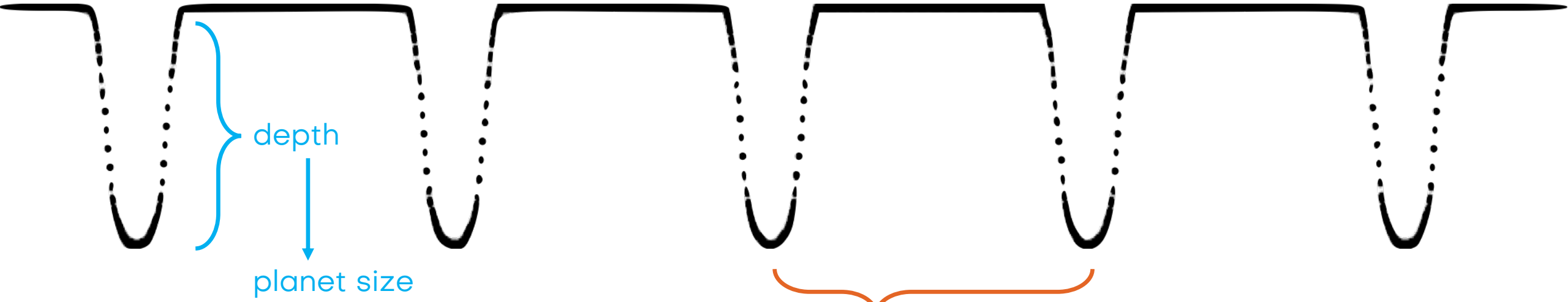


if you watch a star for a long time, you'll see multiple dips



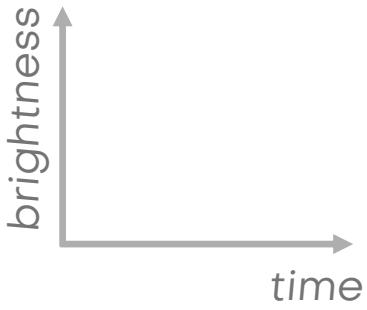
The Transit Method

$$\text{depth} = \left(\frac{R_p}{R_\star}\right)^2$$



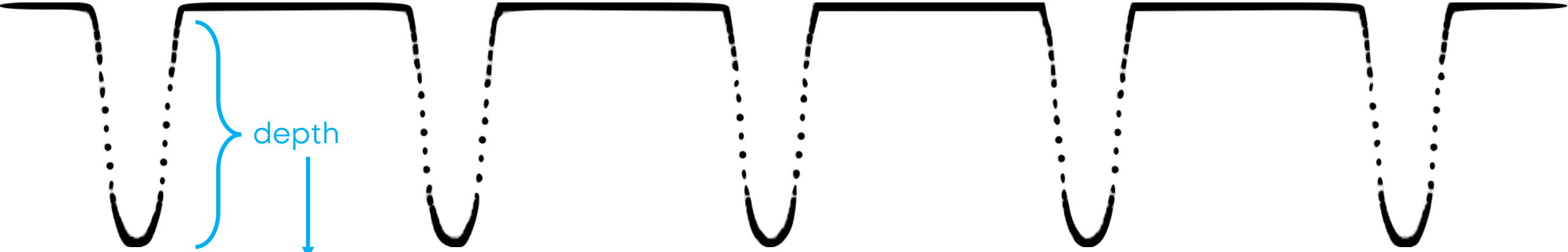
period → length of "year" → distance from star

$$P^2 = \left(\frac{4\pi^2}{GM}\right) a^3$$



The Transit Method

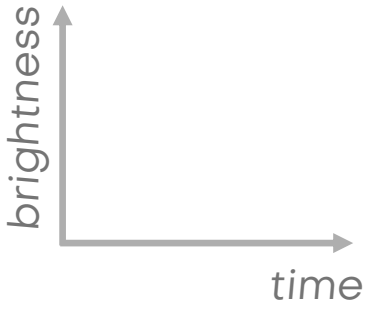
$$\text{depth} = \left(\frac{R_p}{R_\star}\right)^2$$



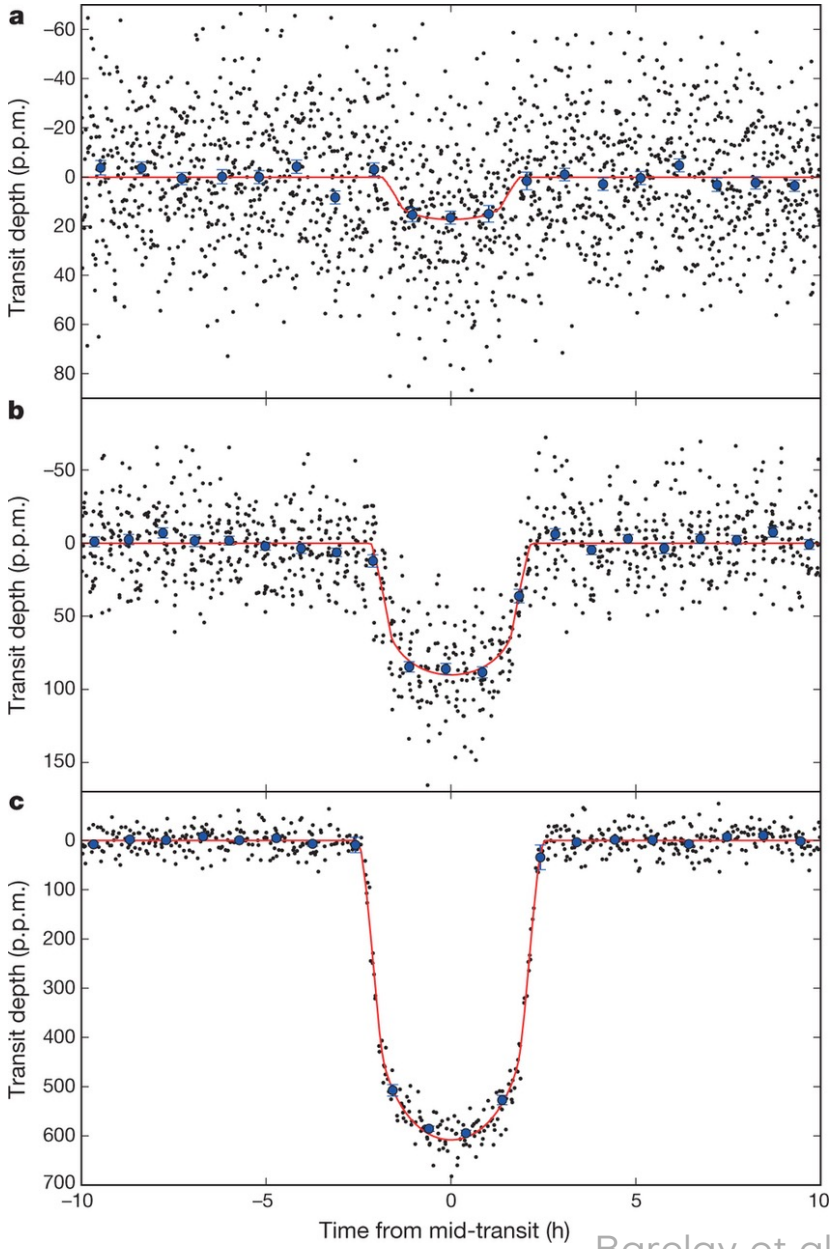
planet density

$$\rho_p = \frac{3M_p}{4\pi R_p^2}$$

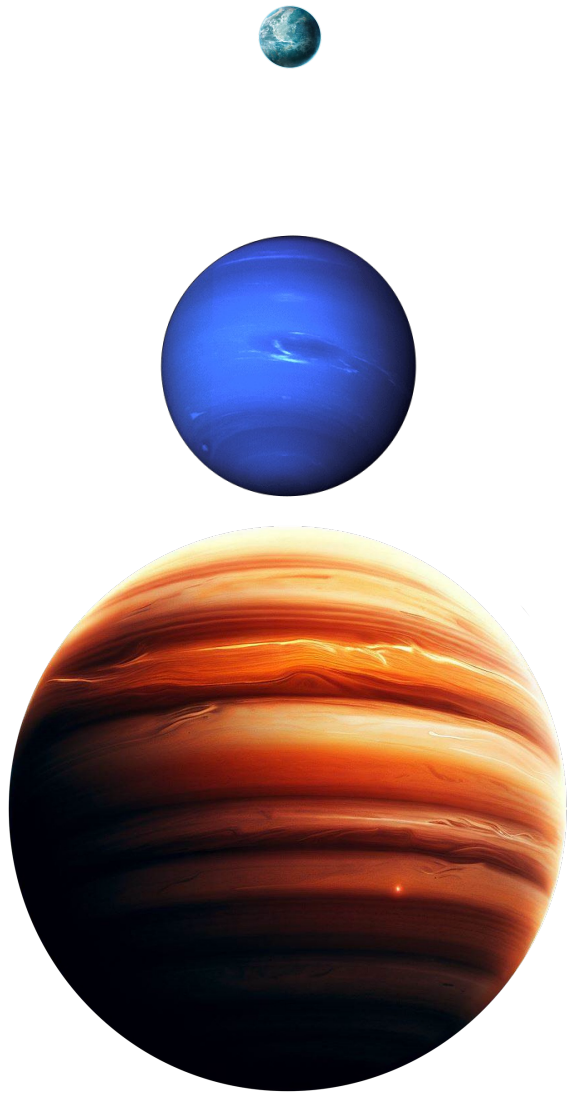
$$P^2 = \left(\frac{4\pi^2}{GM}\right) a^3$$



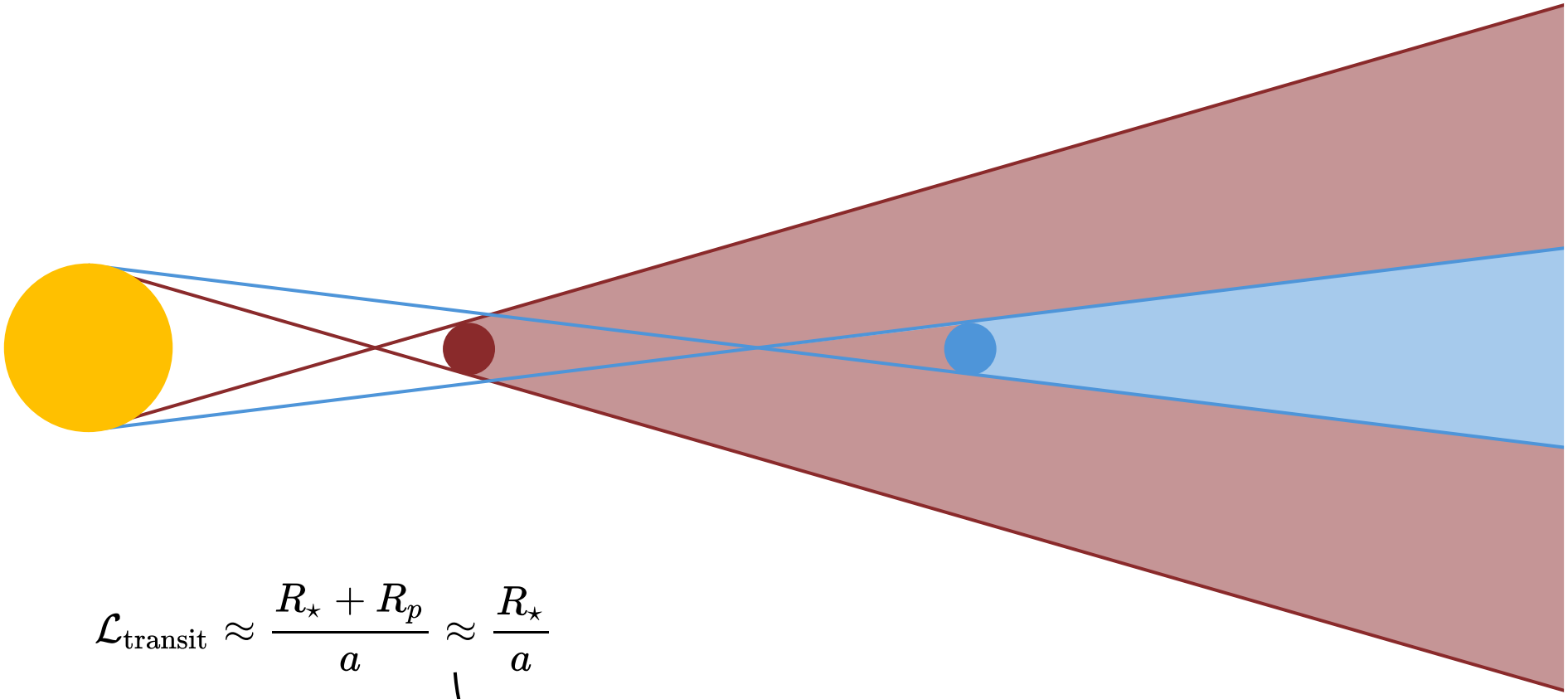
The Transit Method



Barclay et al. (2013)



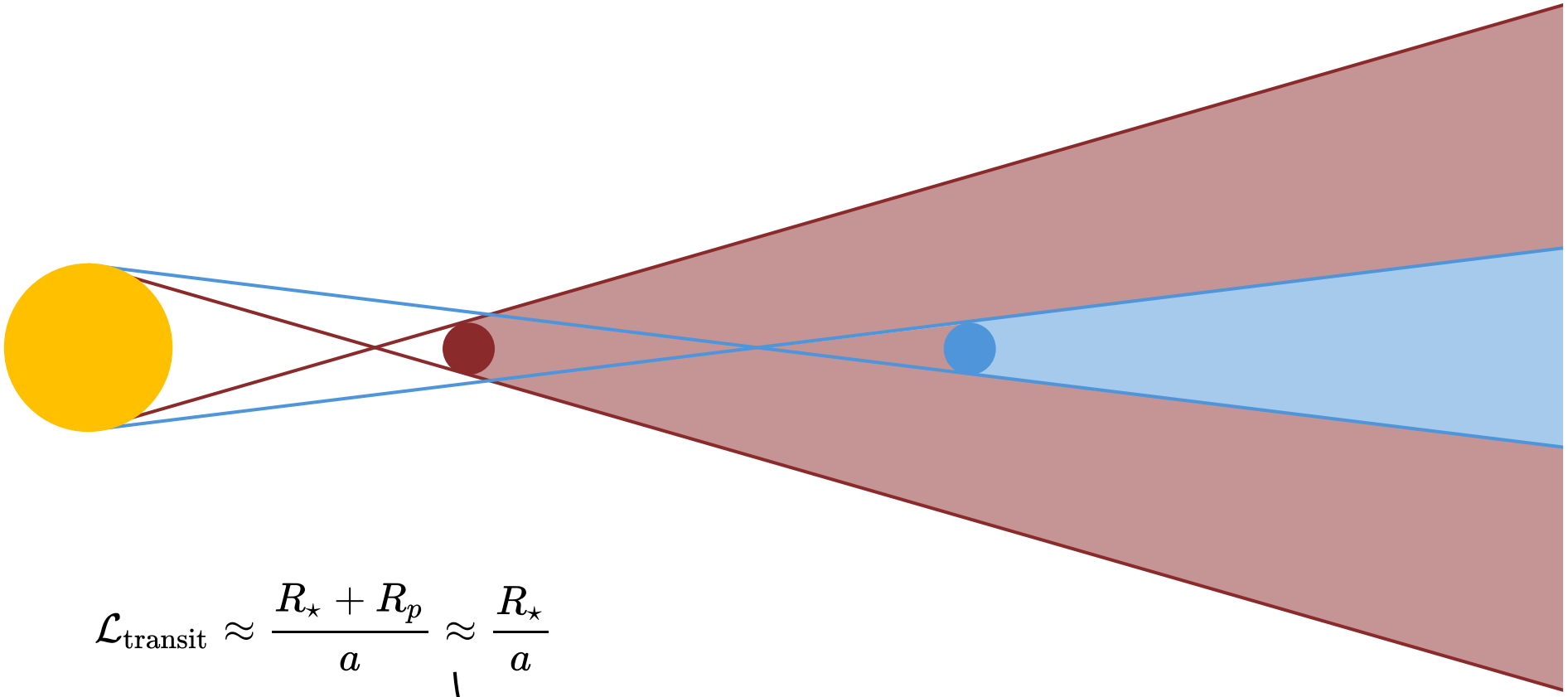
Geometric Transit Likelihood



$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star + R_p}{a} \approx \frac{R_\star}{a}$$

assuming $R_p \ll R_\star$

Geometric Transit Likelihood



one planet!

two planets!

one planet!

no planet :(

$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star + R_p}{a} \approx \frac{R_\star}{a}$$

assuming $R_p \ll R_\star$

The Transit Method

searching for periodic dips in the brightness of stars
caused by a planet blocking light once per orbit

$$\text{depth} = \left(\frac{R_p}{R_\star}\right)^2$$

$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star}{a}$$

best at finding

small (terrestrial) **or** **large** (gas giant)

planets on

short (days) **or** **long** (years)

orbits.

The Transit Method

searching for periodic dips in the brightness of stars
caused by a planet blocking light once per orbit

$$\text{depth} = \left(\frac{R_p}{R_\star}\right)^2$$

$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star}{a}$$

best at finding

small
(terrestrial)

or

large
(gas giant)

planets on

short
(days)

or

long
(years)

orbits.

The Transit Method

searching for periodic dips in the brightness of stars
caused by a planet blocking light once per orbit

$$\text{depth} = \left(\frac{R_p}{R_\star}\right)^2$$

$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star}{a}$$

best at finding

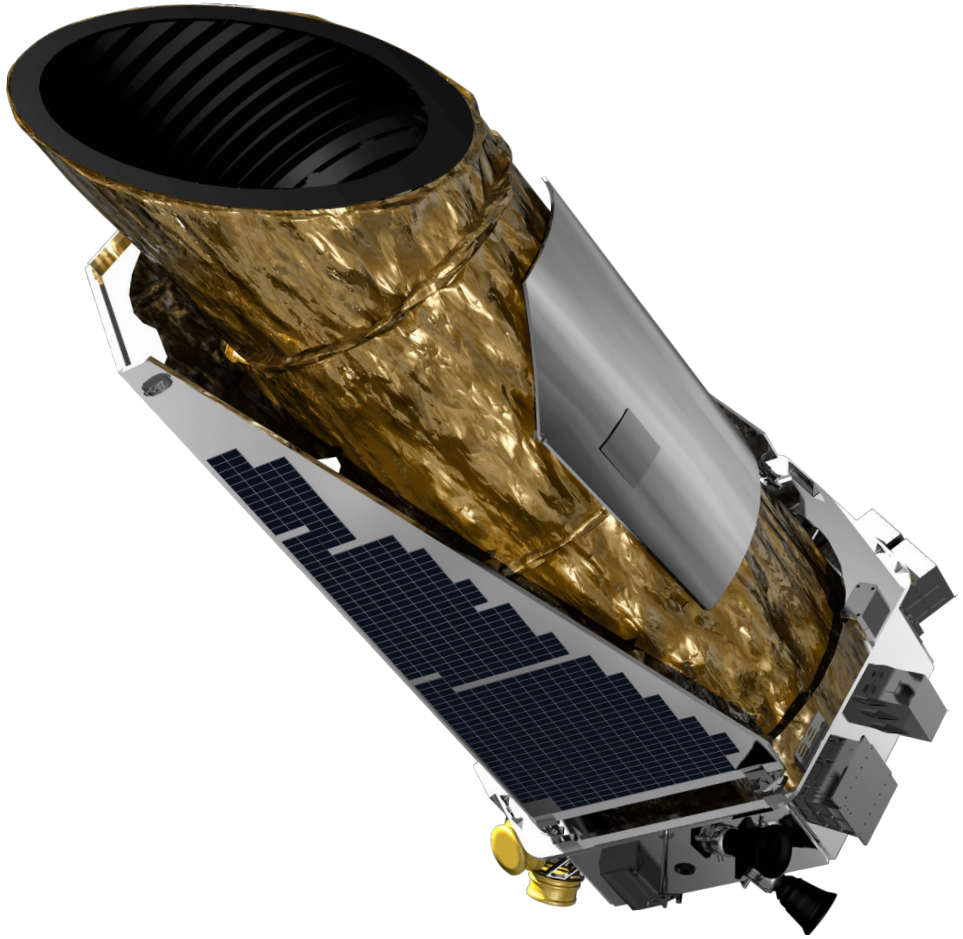
small (terrestrial) **or** **large** (gas giant)

planets on

short (days) **or** **long** (years)

orbits.

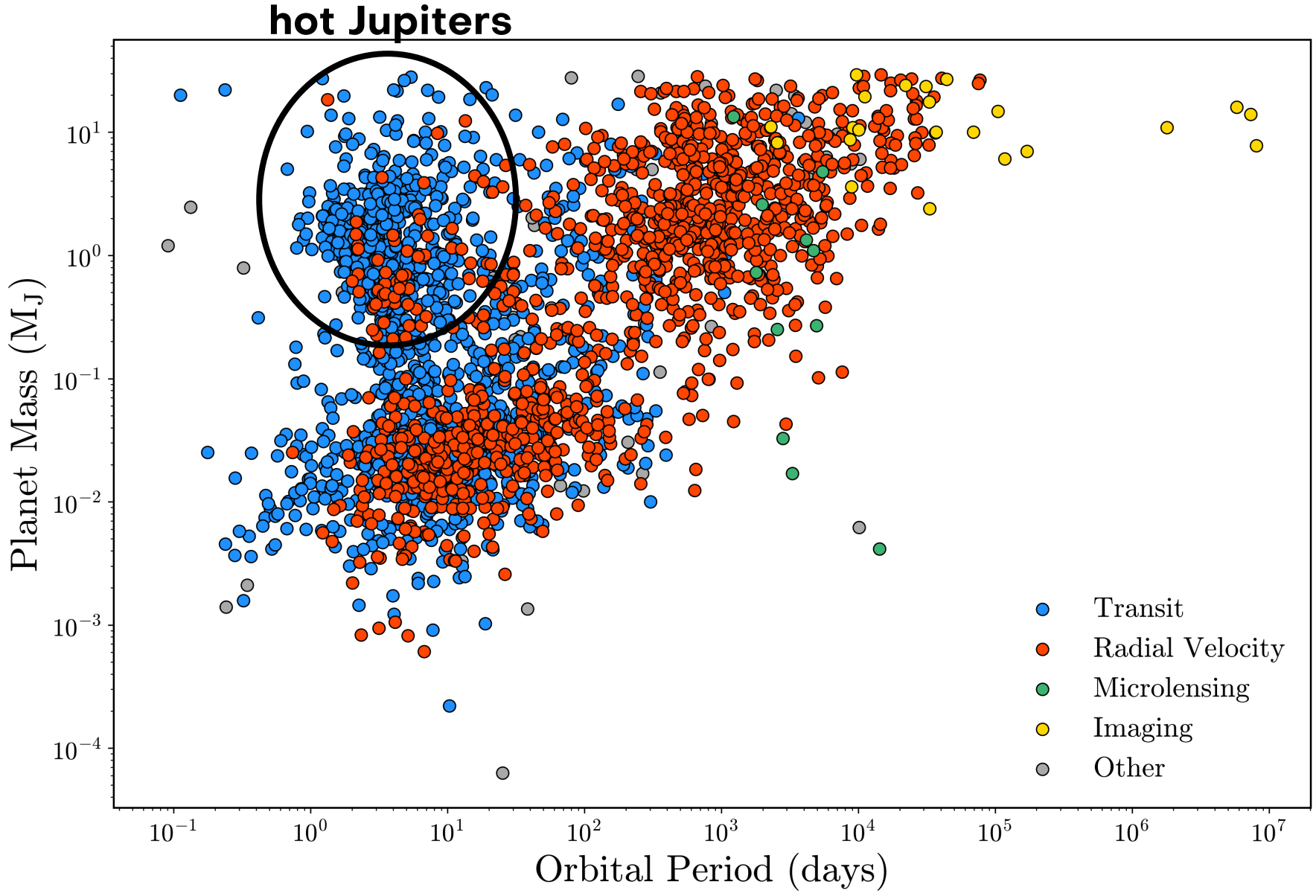
Kepler
2009 – 2018



TESS
2018 – present

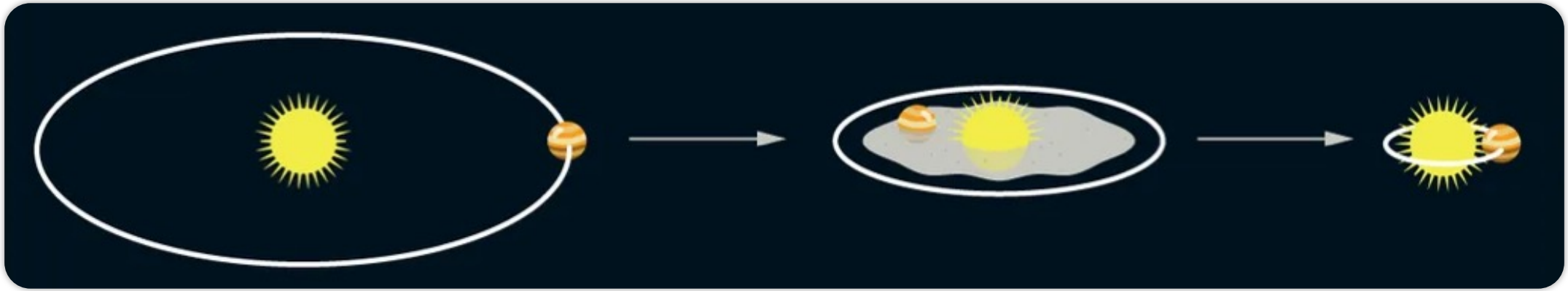


Images: NASA Goddard Space Flight Center

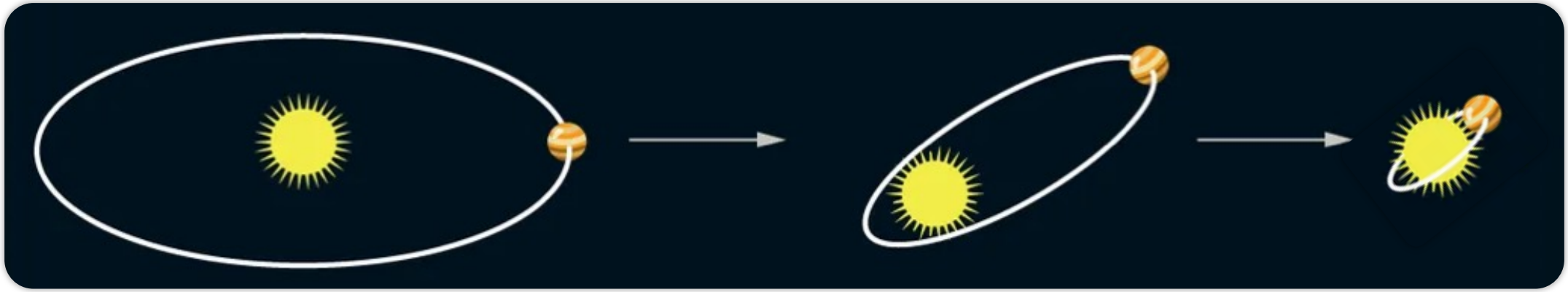

hot Jupiters
Planet Mass (M_J)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
Orbital Period (days)
Transit
Radial Velocity
Microlensing
Imaging
Other

Hot Jupiter Formation Pathways

Disk-driven migration



High-eccentricity migration



Adapted from Dawson & Johnson (2018)

hot Jupiters are unlike anything we see in our solar system



Image: ESO / L. Calçada

hot Jupiters are unlike anything we see in our solar system

...so why do we find so many of them?

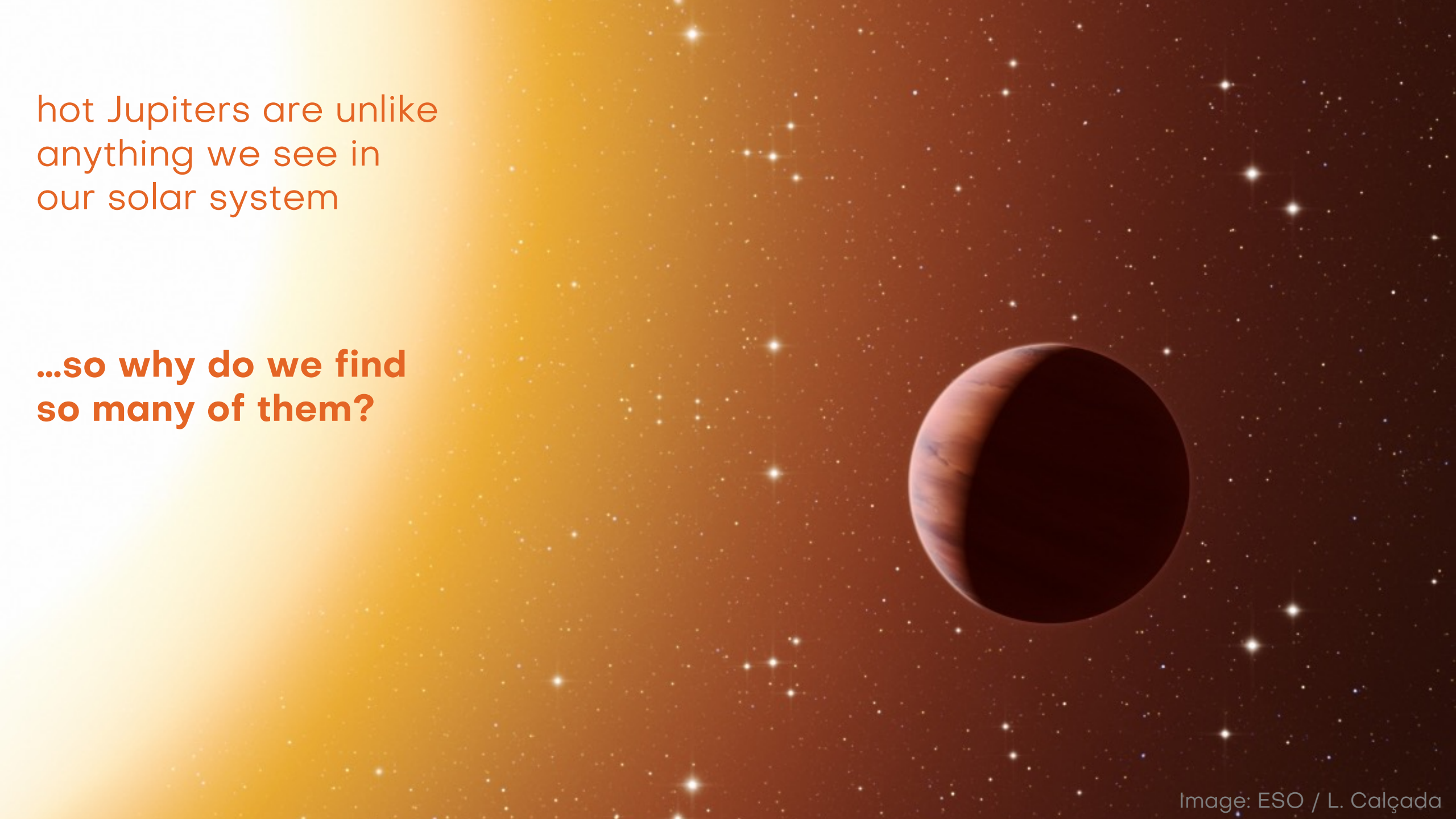


Image: ESO / L. Calçada

Are hot Jupiters common?

there are many in our galaxy, but they're not as common as they appear on this plot

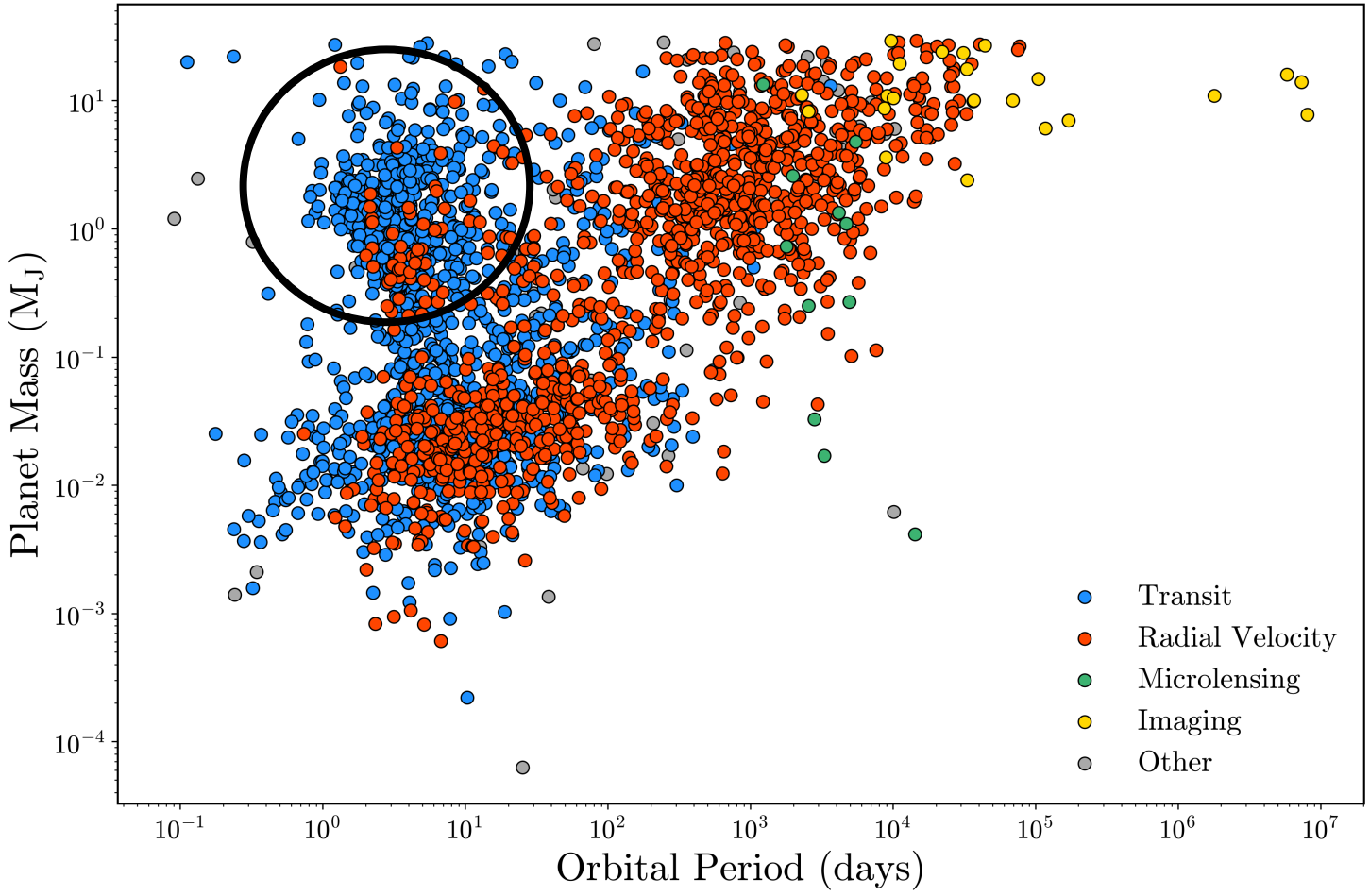




Image: P. Schmidt (Terranaut)

Tutorial: Recovering a transiting planet in TESS data

Step 0:
⟶ Install lightkurve & plot a sample light curve

Step 1:
⟶ Explore the NASA Exoplanet Archive, then make cuts to choose your planet
⟶ Download & plot its TESS light curve

Step 2:
⟶ Find the orbital period using a Box Least Squares periodogram
⟶ Fold the light curve to this period & plot

Step 3:
⟶ Tune the transit depth parameter to estimate the planet radius*
**Requires the star's radius! Look it up on the archive*

Step 4:
⟶ Compare your period & radius to the Exoplanet Archive values
⟶ Plot your planet in context, then submit your results to the class plot

Link: ***nksaunders.space/tutorials***
Make sure to click File → Save a copy in Drive